

M. Sc. Physics (Semester-II) 2026

Quantum Mechanics II – Tutorial 6

1. (a) For two-level quantum system with orthonormal basis $\{|1\rangle, |2\rangle\}$, write its Hamiltonian with respective local energies E_1 and E_2 . (Neglect tunneling between the levels)
- (b) Suppose that a time-dependent perturbation $V(t) = \gamma e^{i\omega t} |1\rangle\langle 2| + h.c.$ is applied to the system at $t > 0$. If the system at $t = 0$ is in state $|1\rangle$, then find the probability that it will be found in each of the two basis states.
- (c) Find the condition under which the probability of being in the state $|2\rangle$ is maximum.
- (d) Plot the corresponding probabilities as a function of time t for frequencies $\omega = \frac{E_2 - E_1}{\hbar}$.

Note: The above problem can be solved exactly. Here we try to solve them using time-dependent perturbation theory.

2. For a one-dimensional (x) harmonic oscillator with characteristic frequency ω , the system is prepared in the ground state $|0\rangle$ at time $t = -\infty$. This is subjected to a perturbation of the form $V_p(x, t) = V_0 \hat{x} e^{-t^2/\tau^2}$, where $\hat{x}$ is position operator.
 - (a) Plot $V_p(x, t)$ as function of time t .
 - (b) What is the probability of finding the oscillator in the excited state $|1\rangle$ at $t = \infty$.
 - (c) Find the value of τ (pulse duration) for which this probability becomes largest for a given transition frequency ω .

(Note that Maximum excitation occurs when the pulse bandwidth matches the transition frequency, i.e., there is an optimal pulse duration that maximizes the excitation probability.)
3. Consider a one-dimensional simple harmonic oscillator with classical angular frequency ω_0 . For $t < 0$ the system is in the ground state. For $t > 0$ a time-dependent perturbation is applied

$$V(t) = F_0 \hat{x} \cos(\omega t)$$

Using time-dependent perturbation theory to lowest nonvanishing order, obtain the expectation value $\langle x(t) \rangle$. Discuss whether the procedure remains valid for $\omega \approx \omega_0$.